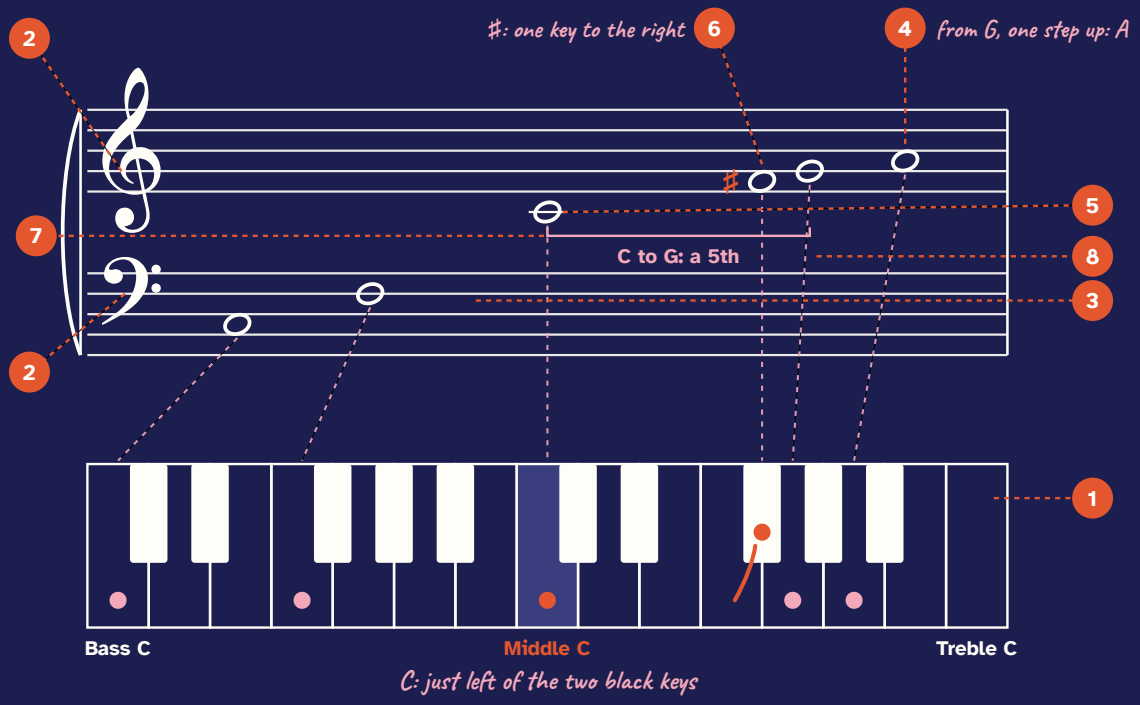
PART I · WHICH NOTE

Pitch

High or low? That is the first question every note answers. Start with the keys under your fingers, find where they live on the staff, see how sharps and flats move them, then measure how far apart two notes are. Everything in this part is in this one picture.



1 **The keyboard** p. 22
two and three black keys; C is left of the two

2 **The staff and the clefs** p. 27
the treble clef names G, the bass clef F

3 **The staff and the keyboard** p. 29
every line and space is one white key

4 **Reading the notes** p. 32
start from a clef note, then step or skip

5 **Ledger lines** p. 36
step out from the staff; Middle C has its own

6 **Sharps, flats and naturals** p. 39
a sharp moves one key right, a flat one key left

7 **Counting intervals** p. 42
count the notes, both ends: C to G is a 5th

8 **Major, minor and perfect** p. 45
count the keys between: 6 make a perfect 5th

★ **Tips: when a note reads wrong** p. 48

PART I · PITCH · 1

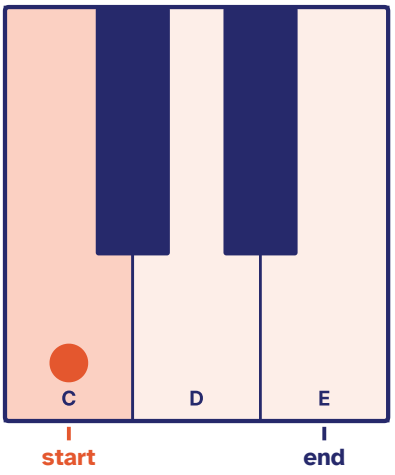
The keyboard

Left is **low**, right is **high**. The black keys come in groups of **two** and **three**. Each group has a first white key, just to its left: **C** for the two, **F** for the three. Find the group, find its first key, then walk on.

THE SECRET

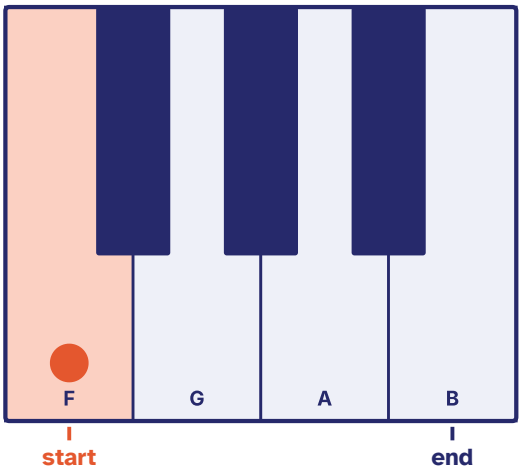
Two black keys: start on C. Three black keys: start on F. Then walk on.

2 black keys: C D E



on the piano E and F touch

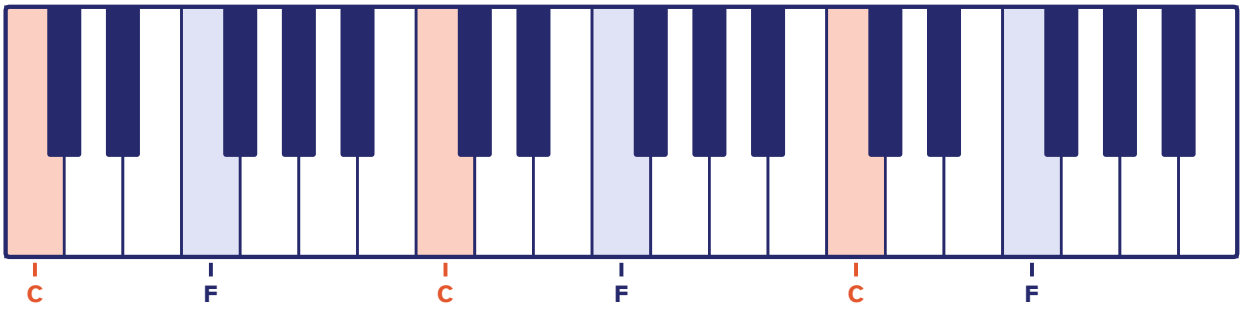
3 black keys: F G A B



Pulled apart: C starts the two, F starts the three · E and B end them

THE RULE

Find the group, then its first key. Looking for D or E? Find two black keys and start on C. Looking for G, A or B? Find three black keys and start on F. You never have to count from far away.

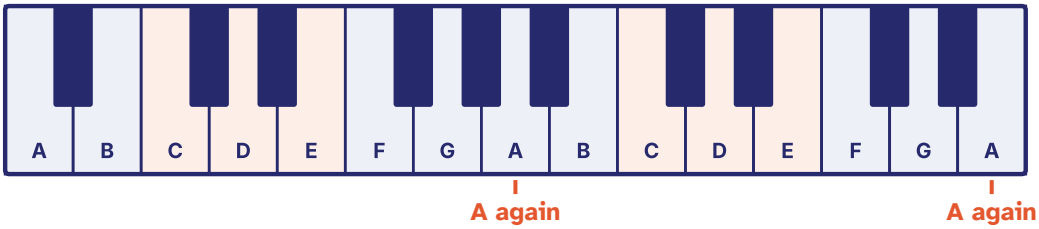


The pattern repeats all the way along: every two starts on C, every three on F

PART I · PITCH · 1 · CONTINUED

Seven letters, round and round

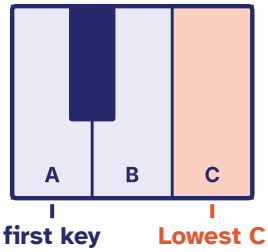
The white keys use only seven letters: **A B C D E F G**. There is no H. After G the next key is **A** again, and the seven letters go round and round, like the days of the week, all the way up the keyboard. **C D E** always sit round the two black keys, **F G A B** round the three. **E–F** and **B–C** have no black key between.



A B C D E F G, A B C D E F G, A …: after G, back to A

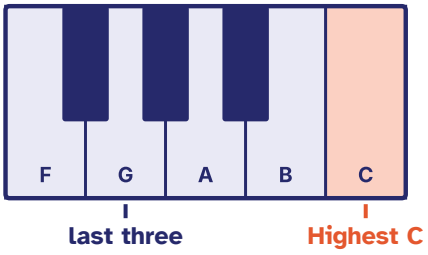
Where the keyboard starts

The very first key, bottom left, is an **A**. Look at the black keys there: only **one**, on its own. It is a group of three that has been cut off: only its end is left. The end of F G A B is **A B**, so the piano starts on A. After A and B comes the first full group of two, and its first key is **Lowest C**.



Where it ends

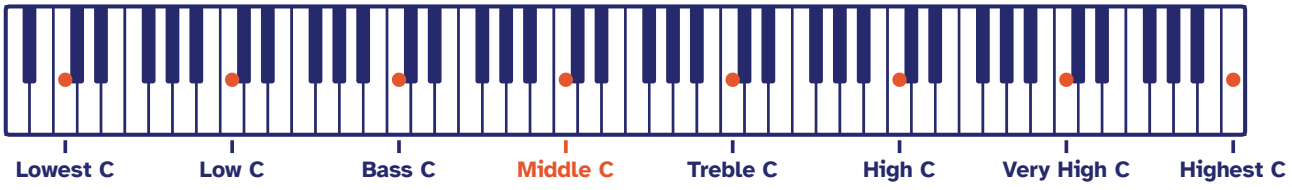
The very last key, top right, has **no black key** after it. Look left: the last group is **three** black keys, F G A B. After B comes C, the first key of a new group of two that was never built. So the last key is a **C**: **Highest C**.



The eight Cs have names

Count the Cs from the bottom and there are eight. The C nearest the middle is **Middle C**. Either side are **Bass C** and **Treble C**; further out, **Low C** and **High C**. At the ends: **Lowest C** at the bottom; at the top, **Very High C**, then **Highest C**, the very last key. (No Very Low C: the piano starts on A.) From one C to the next is an **octave**: C D E F G A B C, eight notes, both Cs counted.

Theory books and apps number them from the bottom instead: C1 is Lowest C, C4 Middle C, C8 Highest C.

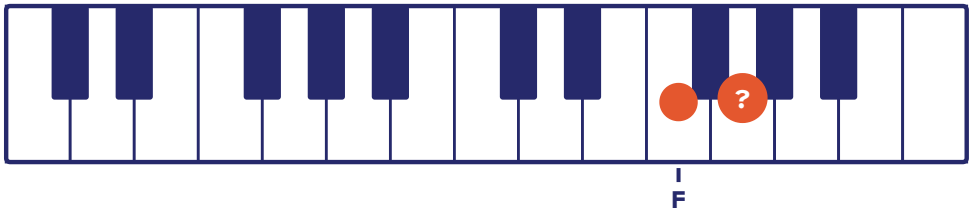


All 88 keys, eight Cs: the five in the middle, Low C to High C, are the ones you will use most

PART I · PITCH · 1 · CONTINUED

EXAMPLE 1

Find G

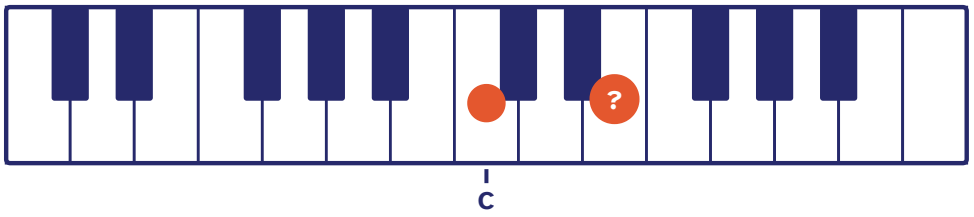


1. **Which family?** G is in F G A B: look for **three** black keys.
2. **Find its first key:** F, just left of the three (the red dot).
3. **Walk on one key.** That is the **?**.

The **?** is **G**: the key after F. No need to start from C and count five.

EXAMPLE 2

Find E

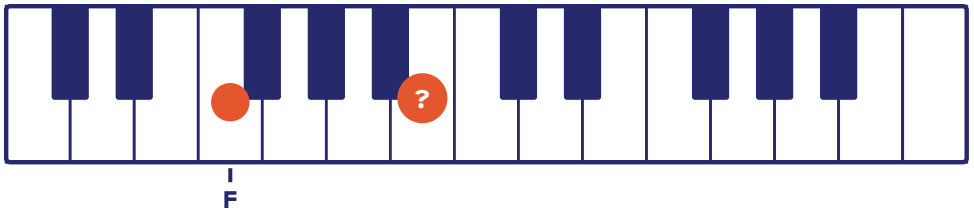


1. **Which family?** E is in C D E: look for **two** black keys.
2. **Find its first key:** C, just left of the two (the red dot).
3. **Walk on:** C, D … the third key is the **?**.

The **?** is **E**: the last of the C D E family, just right of the two black keys.

EXAMPLE 3

Find B



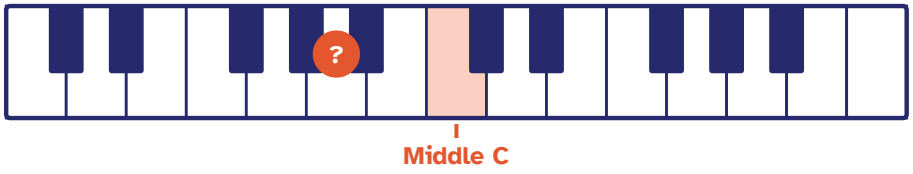
1. **Which family?** B is in F G A B: three black keys.
2. **Start on F** (the red dot).
3. **Walk on:** F, G, A … the last key of the family is the **?**.

The **?** is **B**: the last of F G A B, just right of the three black keys.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 4

Find the A below Middle C

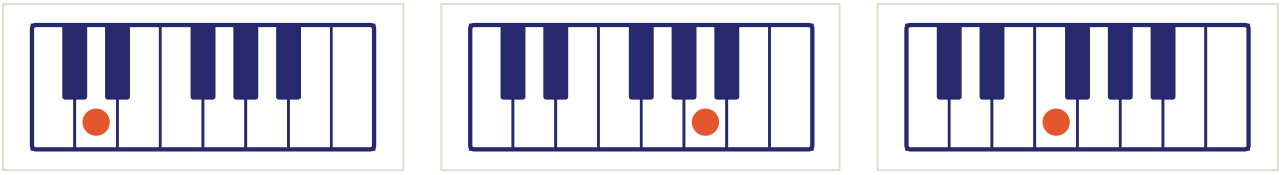


1. **Which family?** A is in F G A B.
2. **From Middle C, look left** for the nearest three black keys, and start on their F.
3. **Walk on:** F, G … the third key is the **?**.

The **?** is **A**: F, G, A. Find the family first, and every key is at most three steps away.

GAME

Name the key



1 2 3

Find the group, then its first key, then walk on. Name each red key.

Answer: 1 D (C, then D) · 2 A (F, G, A) · 3 F (the first of the three).

GAME

Which family?

1	2	3	4	5	6
D	A	F	E	B	G

Does each white key live with the **two** black keys (start on C) or the **three** (start on F)?

Answer: 1 two · 2 three · 3 three (the first) · 4 two (the last) · 5 three (the last) · 6 three.

EXAMPLE 5

Higher or lower?



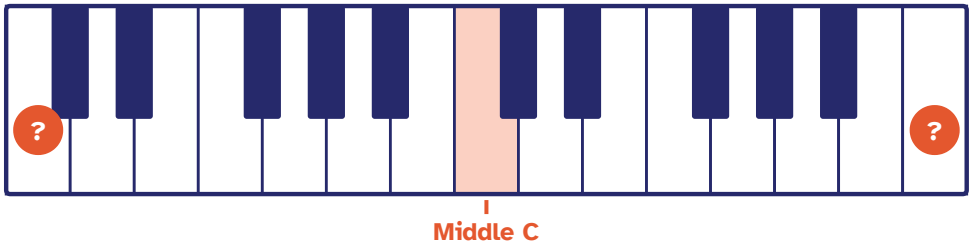
1. **Find the two red keys.**
2. **Which one is further right?**
3. **Right means higher:** play both and listen.

The second note is **higher**: it is to the right. Left is always lower, right always higher.

PART I · PITCH · 1 · CONTINUED

EXAMPLE 6

Find Bass C and Treble C



1. **Start on Middle C.**
2. **Walk left** to the next pair of black keys. The C just left of it is one **?**.
3. **Walk right** the same way, to the next pair: the other **?**.

Left **?**: **Bass C**. Right **?**: **Treble C**. Each is one octave, eight notes, from Middle C.

TRY IT · 10 SECONDS

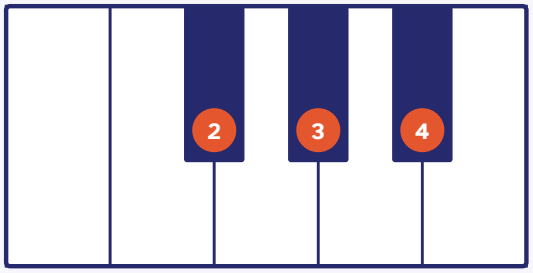


Play every C, then every F.

Bottom to top, saying the names: Lowest C, Low C, Bass C, Middle C, Treble C, High C, Very High C, Highest C. Then every F: left of the three.

YOU KNOW THIS

Hot Cross Buns, on three black keys



Right hand, fingers **2 3 4** on the three black keys (thumb is 1, little finger 5). Play from the top: **4 3 2**, 4 3 2, then quickly, twice as fast, **2 2 2 2, 3 3 3 3**, and 4 3 2 to finish. A whole tune, and not one white key.

Try it. Find another group of three, higher up, and play it again there.

THINK OF IT The black keys are bumps you can feel in the dark. Close your eyes and slide your hand along: two bumps, and the key to their left is C; three bumps, and the key to their left is F.

TRAP

- × Counting every key from C: find the nearest group and start from its C or F.
- × Mixing up the families: C is left of the **two** black keys, F left of the **three**.

SEE ALSO

The staff and the clefs p. 27 · The staff and the keyboard p. 29

PRACTISE IT

Workbook Pitch 1 · Online: Keyboard Patterns · The Musical Alphabet

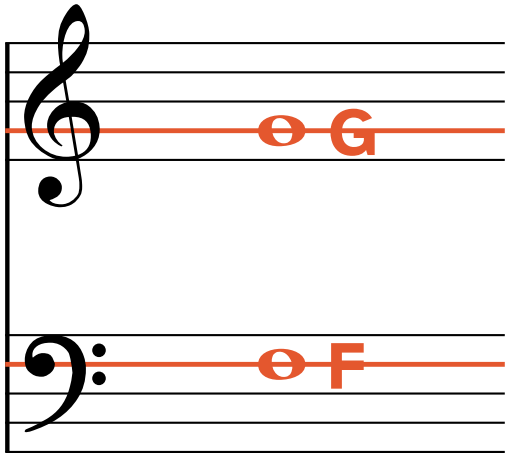
PART I · PITCH · 2

The staff and the clefs

Music is written on a **staff**: five lines and four spaces, counted from the **bottom**. Higher on the staff means higher on the keyboard. The **clef** at the start is the key to the code: it names one line, and every other note follows from it.

THE SECRET

The treble clef curls round G; the bass clef's two dots hug F.





Treble clef = G clef
curls round line 2: G

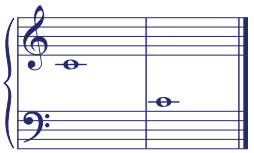


Bass clef = F clef
its dots hug line 4: F

The treble clef is the G clef: it curls round G · the bass clef is the F clef: its dots hug F

The grand staff

Piano music uses two staves joined by a brace: the **treble** staff on top, usually for the right hand, and the **bass** staff below, usually for the left. **Middle C** sits between them on a short line of its own, and can be written in either staff.

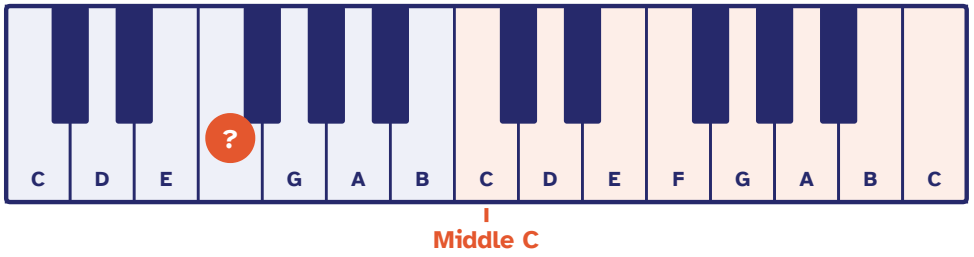


Middle C in the treble staff (left) and in the bass staff (right): the same key.

PART I · PITCH · 2 · CONTINUED

EXAMPLE

Which staff is this key written on?



1. **Find Middle C.**
2. **Is the ? to its left or right?** Left is lower: the blue half.
3. **Low keys go on the bass staff**, high keys on the treble staff.

The **?** is below Middle C, so it is written on the **bass staff**, usually for the left hand.

GAME

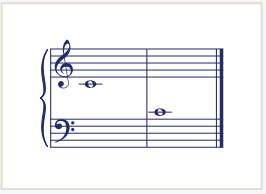
Treble or bass?

1. the G above Middle C
2. the lowest key on the piano
3. Middle C
4. High C

Which staff would each note be written on?

Answer: 1 treble · 2 bass · 3 either: it sits between the two · 4 treble.

TRY IT · 10 SECONDS



Point, then play.

Point to the line the treble clef curls round and play that G, the G just above Middle C. Point to the line between the bass clef's dots and play that F, the F just below Middle C.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The treble clef curls round **line 2: G**, the very note Twinkle leaps up to. Read the round head, not the stem: the stem never changes the note.

Try it. Point to the G line, then play C C G G A A G.

THINK OF IT A clef is a pair of glasses: the same five lines read as different notes, depending on which pair you put on.

TRAP

× Reading bass-clef notes as if they were treble: check the clef on every line.

SEE ALSO
The staff and the keyboard p. 29 · Reading the notes p. 32

PRACTISE IT
Workbook Pitch 2 · Online: The Staff & the Two Clefs

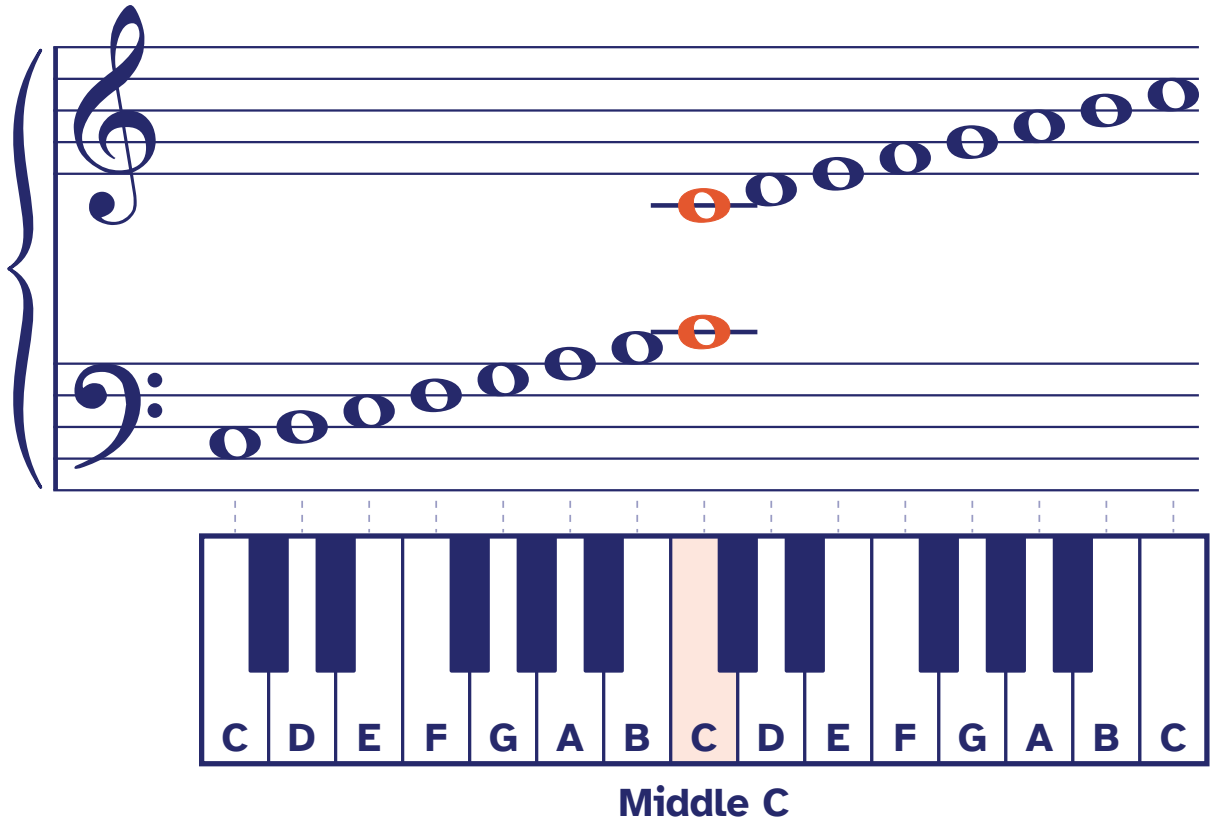
PART I · PITCH · 3

The staff and the keyboard

Each line and each space of the grand staff is one white key. Climb the staff and you climb the keyboard: here are two octaves, from Bass C to Treble C, with Middle C in red where the two staves meet.

THE SECRET

Every line and every space is one white key, in order.



Between the two staves there is only one key: **Middle C**, written twice, below the treble staff or above the bass staff. No other key is missed out.

THE RULE

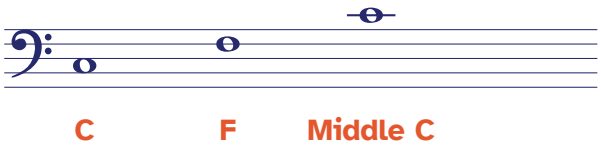
One step on the staff (line to space, or space to line) is the very next white key. The black keys sit in between; how they are written is chapter 6 (p. 39).

Up the staff is to the **right** on the keyboard; down the staff is to the **left**. A note on a line and the note in the space just above it are next-door white keys, so no key is ever missed out.

PART I · PITCH · 3 · CONTINUED

EXAMPLE 1

From the page to the key

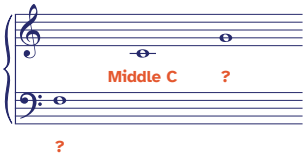


1. **Name the note:** line 4 of the bass staff, between the clef's dots, is F.
2. **On the piano,** find the three black keys just below Middle C.
3. **F is the first key** of that family: play it with the left hand.

The bass-clef F is the **F just below Middle C**: left of the three black keys under it.

EXAMPLE 2

Mirror notes either side of Middle C

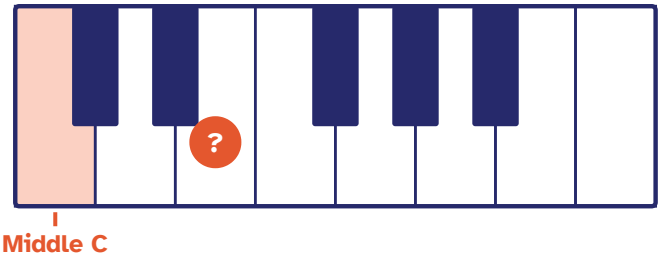


1. **Treble G:** the three black keys just above Middle C; G is the second key, after F.
2. **Bass F:** the three black keys just below Middle C; F is the first key.
3. **Now count each from Middle C:** C D E F G up, C B A G F down. Which number?

Both are **note 5**: mirror images. Fold the grand staff at Middle C and treble G lands exactly on bass F; the staves meet line for line.

EXAMPLE 3

From the key to the page



1. **Find the ? key:** count up from Middle C, C 1, D 2 ...
2. **Do the same on the staff:** Middle C on its ledger line, D in the space under the staff ...
3. **Where do you land?**

The **?** is **E**, note 3, and it sits on the **bottom line** of the treble staff.

PART I · PITCH · 3 · CONTINUED

GAME

Count the notes

1	2	3
Middle C up to A	Middle C down to E	G up to the next G

Count the notes, both ends included. The staff climbs the same way: one line or space per note.

Answer: 1 six (C D E F G A) · 2 six (C B A G F E) · 3 eight: an octave.

TRY IT · 10 SECONDS

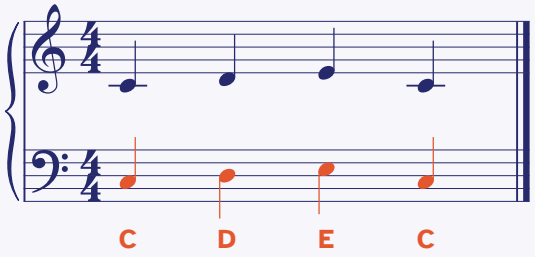


Play Middle C, then walk up.

Say each white key as you play it, C D E F G A B C, and picture each one climbing the staff: line, space, line, space.

YOU KNOW THIS

Frère Jacques, in both hands



The left hand plays the **same letters**, C D E C, one octave lower, written in the bass clef.

Try it. Play it right hand, then left hand, then both together.

THINK OF IT The staff is a ladder laid beside the keyboard: each rung, a line or a space, is the next white key.

TRAP

- × Skipping a key: every line and every space is used, one white key each.
- × Counting from 0: the note you start on is note 1.

SEE ALSO

The staff and the clefs p. 27 · Reading the notes p. 32

PRACTISE IT

Workbook Pitch 3 · Online: The Staff & the Two Clefs · Line and Space Names

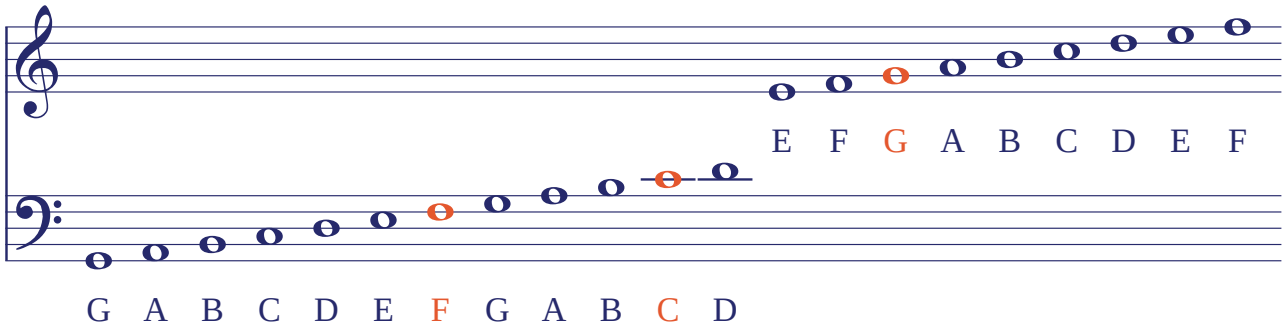
PART I · PITCH · 4

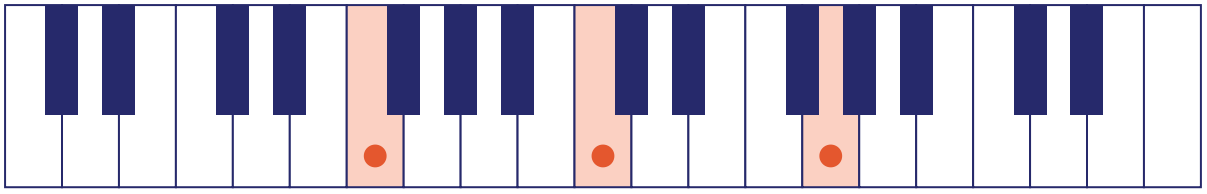
Reading the notes

Every line and space has one letter. You do not have to learn them all at once: start from a note you always know, then **step** or **skip** to the one you want.

THE SECRET

Start from a note you know (bass F, Middle C, treble G), then step or skip.





Middle C in the middle; treble G and bass F, the clef notes, either side. Every line and space is the next letter, with its own key underneath (the keyboard starts on G, so its first black keys belong to a group of three).

THE RULE

Step: to the very next line or space (one letter on). Skip: line to the next line, or space to the next space (two letters on).

Line note or space note?

A **line note** has the line running through its middle. A **space note** sits between two lines and touches both. Count both from the bottom: line 1 is the bottom line, space 1 the bottom space.

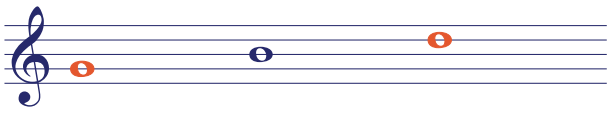
Words to remember the lines and spaces

STAFF	LINES (BOTTOM TO TOP)	SPACES (BOTTOM TO TOP)
Treble	**E G B D F**: *Every Good Boy Deserves Fun*	**F A C E**: they spell *FACE*
Bass	**G B D F A**: *Great Big Dragons Fly Away*	**A C E G**: *All Cows Eat Grass*

PART I · PITCH · 4 · CONTINUED

EXAMPLE 1

Read a note you are not sure of

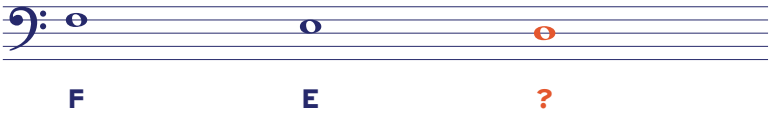


1. **Treble lines** from the bottom: *Every Good Boy Deserves Fun*.
2. **The other red note, on line 4:** the fourth word. Which word, which letter?
3. **Check it** from the clef note: G on line 2, skip to B, skip to …

Line 4 is *Deserves*: **D**. And from the clef: G, B, D, two skips up.

EXAMPLE 2

Read a bass note from the clef



1. **Bass lines** from the bottom: *Great Big Dragons Fly Away*.
2. **The red note is on line 3:** the third word. Which letter?
3. **Check it** from the clef note: the dots hug F on line 4; step down to E, then …

Line 3 is *Dragons*: **D**. And from the clef: F, E, D, two steps down.

EXAMPLE 3

One line, two notes



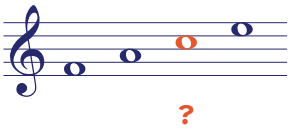
1. **Look at the red notes:** both sit on the bottom line.
2. **Now look at the clefs:** treble on the left, bass on the right.
3. **Use each clef's sentence:** the first word of the treble lines, *Every…*; the first word of the bass lines, *Great…*.

Treble bottom line: **E** (*Every*). Bass bottom line: **G** (*Great*). Same line, different note: always read the **clef** first.

PART I · PITCH · 4 · CONTINUED

EXAMPLE 4

Find Treble C on the staff

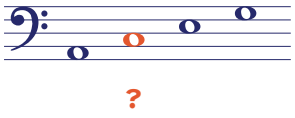


1. **These notes sit in the spaces.** The bottom space is F.
2. **Step up through the spaces:** F, A … they spell a word.
3. **The red note is space 3.** Which letter is it?

F A **C** E: space 3 is **C**, Treble C, the same C you found on the keyboard, one octave above Middle C.

EXAMPLE 5

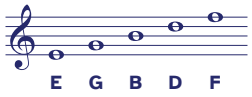
Find Bass C on the staff



1. **These notes sit in the bass spaces.** The bottom space is A: *All.*
2. **Say the next word:** *All Cows …*
3. **The red note is space 2.** Which letter?

A **C** E G: space 2 of the bass staff is **C**, Bass C, one octave below Middle C.

TRY IT · 10 SECONDS



Say the treble lines up and down.

"E G B D F", then "F D B G E", out loud, as fast as you can without a slip. Then the spaces: "F A C E".

GAME

Name it from the clef note

1	2	3	4
treble line 3	treble space 1	bass line 3	bass space 3

Start from the clef note (treble G on line 2, bass F on line 4), then step or skip.

Answer: 1 B (skip up from G) · 2 F (step down from G) · 3 D (two steps down from F) · 4 E (step down from F).

PART I · PITCH · 4 · CONTINUED

HOW TO

Read a new note in three moves

1. **Look at the clef**: treble or bass? That decides every letter.
2. **Find the nearest note you know**: treble G, bass F or Middle C, or use the sentence for lines or spaces.
3. **Step or skip** to the new note, say its name, then play it.

EXTRA · GOING FURTHER

Read the shape, not just the notes

Good readers see the jumps before the letters. Notes that touch, line then space, are a **step**: the next letter. Notes with one line between them, line to line or space to space, are a **skip**: one letter missed out. Name the first note, then let the shape name the rest: up a step, up a skip, down a step.

YOU KNOW THIS

Mary Had a Little Lamb



E D C: **line, space, Middle C's own little line**, one step down each time. Read the shape and the tune walks down with it.

Try it. Read the notes aloud before you play them.

THINK OF IT The clef notes are lamp posts on a dark road: walk from the nearest one and you are never lost.

TRAP

- × Naming bass notes with treble names: in the bass, the bottom line is G, not E.
- × Mixing up step and skip: a step changes line to space; a skip stays on lines (or on spaces).
- × Mistaking a space note for a line note: a line runs through a line note; a space note sits between two lines.
- × Counting lines from the top: line 1 is always the bottom line.

SEE ALSO

The staff and the keyboard p. 29 · Ledger lines p. 36

PRACTISE IT

Workbook Pitch 4 · Online: Line and Space Names · Steps & Skips

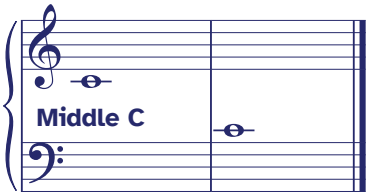
PART I · PITCH · 5

Ledger lines

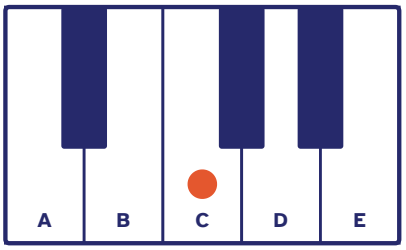
Some notes are too high or too low for the staff. They sit on short extra lines called **ledger lines**, spaced exactly like the staff lines. Read them the way you read the staff: step out from the nearest staff line, line, space, line.

THE SECRET

Step out from the nearest staff line: line, space, line. The first ledger line is the one nearest the staff.

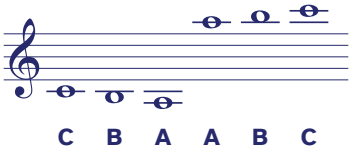


Middle C

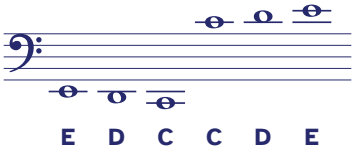


Middle C sits on a ledger line of its own: below the treble staff, or above the bass staff. One key, two ways to write it.

Below and above the staff



Treble: C, B, A below; A, B, C above.



Bass: E, D, C below; C, D, E above.

Name ledger notes by stepping out from the staff, one letter at a time. Treble: below the bottom line E come D, then Middle C on the first ledger line, then B and A. Above the top line F come G, A on the first ledger line, B, then C on the second. Bass: below the bottom line G come F, E on the first ledger line, D, C; above the top line A come B, then Middle C. The few keys round Middle C, from A below it to E above it, can be written in either staff: the hand that plays them decides.

THE RULE

Count ledger lines from the staff outwards. The first ledger line is the one nearest the staff. Step out line, space, line, and never skip one.

PART I · PITCH · 5 · CONTINUED

EXAMPLE 1

Read a high note on ledger lines



1. **The top line** of the treble staff is F.
2. **Step up** one note at a time: G sits on top of the staff, A on the first ledger line.
3. **Keep stepping** to the red note: B above it, then the second ledger line.

Two ledger lines above the treble staff: **High C**, two octaves above Middle C.

EXAMPLE 2

Read a low note on ledger lines

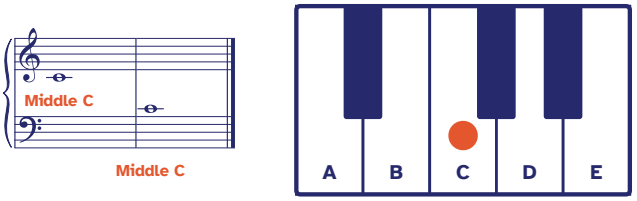


1. **The bottom line** of the bass staff is G.
2. **Step down** one note at a time: F in the space below, E on the first ledger line.
3. **Keep stepping** to the red note: D below it, then the second ledger line.

Two ledger lines below the bass staff: **Low C**, two octaves below Middle C. Its mirror is High C, two ledger lines above the treble.

EXAMPLE 3

Middle C, two ways



1. **Treble staff:** Middle C hangs on the first ledger line below.
2. **Bass staff:** Middle C sits on the first ledger line above.
3. **Find both on the keyboard.** Are they different keys?

The same key, **Middle C**: the right hand reads it in the treble, the left hand in the bass.

PART I · PITCH · 5 · CONTINUED

The four first ledger lines

WHERE	FIRST LEDGER LINE	NOTE
Treble, below the staff	just under line 1	**Middle C**
Bass, above the staff	just over line 5	**Middle C**
Treble, above the staff	just over line 5	**A**
Bass, below the staff	just under line 1	**E**

Two of them are the same key, Middle C. The other two are not: always look at the clef first.

EXTRA · GOING FURTHER

Too many ledger lines? 8va

Very high notes need lots of ledger lines, and lots of ledger lines are hard to read. So a high passage is often written lower, with **8va** and a dotted line above it: play those notes **one octave higher** (eight notes up, from a C to the next C, p. 23) until the dotted line stops. **8vb** under the notes means an octave lower. (On its own, 8ve is just short for octave.)

GAME

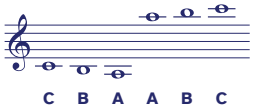
Name the ledger note

1. treble, first ledger line below
2. bass, first ledger line above
3. treble, first ledger line above
4. bass, first ledger line below

Step out from the nearest staff line.

Answer: 1 Middle C · 2 Middle C · 3 A · 4 E.

TRY IT · 10 SECONDS



Say the ledger notes.

Treble: C, B, A below; A, B, C above. Point to each one, say it, then play it.

THINK OF IT Ledger lines are the staff's ladder carried on past its ends: the same rungs, just shorter.

TRAP

- × Reading ledger lines from the wrong end: step out from the staff.
- × Counting from the outside in: the first ledger line is the one nearest the staff.

SEE ALSO
Reading the notes p. 32 · Sharps, flats and naturals p. 39

PRACTISE IT
Workbook Pitch 5 · Online: Ledger Lines

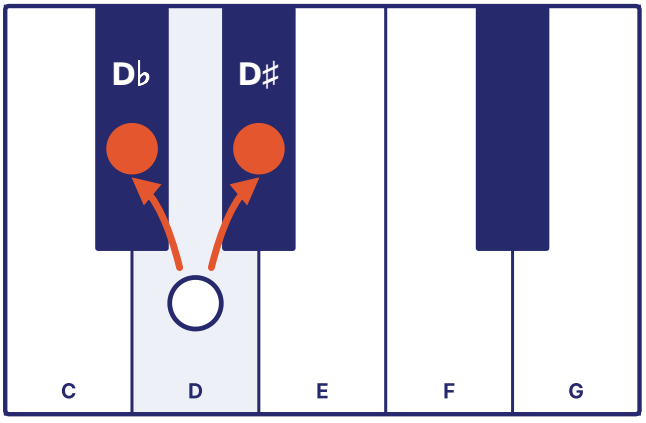
PART I · PITCH · 6

Sharps, flats and naturals

A sign in front of a note moves it to the very next key. **Sharp**: one key to the **right**, a **semitone** higher. **Flat**: one key to the **left**, a semitone lower. Usually that next key is black, which is how the black keys get their names. These signs are called **accidentals**.

THE SECRET

Sharp: the very next key to the right. Flat: the very next key to the left.



♯ one key right · ♭ one key left

SIGN	NAME	WHAT IT DOES	FIRST MET
♯	sharp	one key to the right (a semitone higher)	Level 0
♭	flat	one key to the left (a semitone lower)	Level 0
♮	natural	cancels a sharp or flat: back to the white key	Level ½

A **semitone** is the smallest step on the piano: from one key to the very next, black or white (American books call it a *half step*). More on p. 42. With no black key between (E–F, B–C), the next key is white: E♯ = F and B♯ = C; F♭ = E and C♭ = B. And every black key has two names: C♯ = D♭.

THE RULE

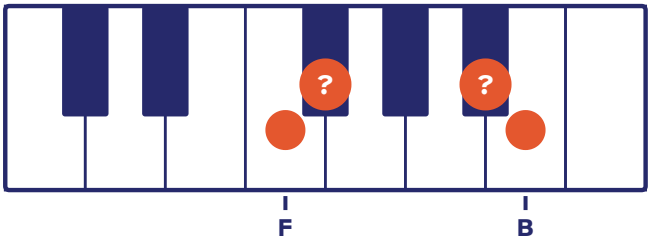
A sign belongs to one line or space. It changes only that line or space, in that octave, and only until the bar line. The same letter in another octave, and every other note, stays as it is.

A **bar** is the space between two **bar lines** (p. 50). A **key signature** at the start of every line is different: it lasts the whole piece (p. 99). And a natural cancels a sharp just as it cancels a flat.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 1

Find F♯ and B♭



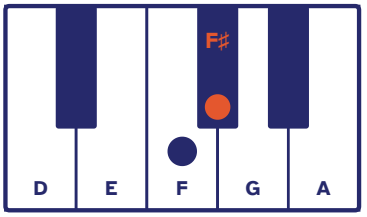
1. **Find F and B** (the red dots): F just left of the three black keys, B just right of them.
2. **Sharp means right:** F♯ is the very next key to the right of F.
3. **Flat means left:** B♭ is the very next key to the left of B. Black or white?

The left **?** is **F♯**, the first black key of the three; the right **?** is **B♭**, the highest one, on the right.

EXAMPLE 2

How long does the sharp last?



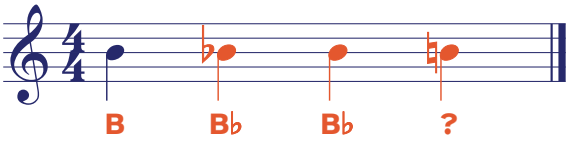


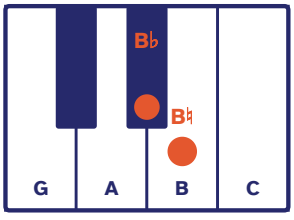
1. **The first note** has a sharp: F♯.
2. **Later in the same bar** there is no sign. Is it still sharp?
3. **After the bar line**: is it still sharp?

Still **F♯** until the bar line; after it, plain **F** again. A sign lasts **to the end of its bar**.

EXAMPLE 3

A natural cancels the flat





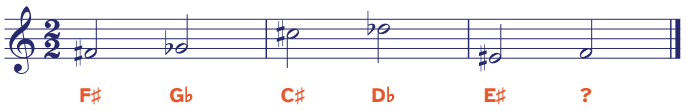
1. **B**: the white key.
2. **B♭**: one key left, and it stays flat for the rest of the bar.
3. **The last note** has a natural sign: what does it do?

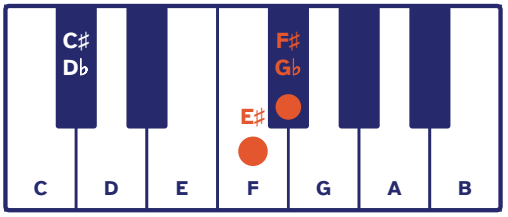
The natural **cancels** the flat: back to the white key B, until the bar line.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 4

One key, two names





1 **The first bar:** F♯, then G♭. Find both on the keyboard.

2 **Are they different keys?**

3 **The last bar:** E♯. Which key is one right of E?

F♯ and G♭ are the **same black key**; E♯ is the white key **F**, so the **?** is plain F, the same key as E♯. Same sound, different names: **enharmonic**.

GAME

Which key is it?

1	2	3	4
C♯	E♭	B♯	F♭

Sharp: one key to the right. Flat: one key to the left. Which key do you press?

Answer: 1 the black key right of C · 2 the black key left of E · 3 the white key C · 4 the white key E.

YOU KNOW THIS

Für Elise (Beethoven)



E, **D♯**, E, **D♯**: the next key down and back. The bar line wipes the sharp out, so it is written again; later the D needs a **natural**, or it would still be D♯.

Try it. Play E and D♯ back and forth: the black key just left of E.

THINK OF IT A sharp or flat is a sticky note on one line of the staff: it stays stuck until the bar line, then it falls off.

TRAP

× Carrying a sharp or flat past the bar line: after it, the note is plain again.

SEE ALSO

Keys and key signatures p. 99

PRACTISE IT

Workbook Pitch 6 · Online: Accidentals · Tones & Semitones

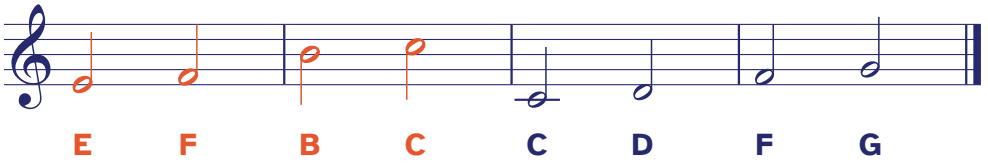
PART I · PITCH · 7

Counting intervals

The smallest step on the piano is a **semitone**: from one key to the very next. Two semitones make a **tone**. Bigger distances are **intervals**: to give one a number, **count the notes**, both ends included.

THE SECRET

The very next key is a semitone; two make a tone. For an interval, count the notes, both ends included.



E–F and B–C: next door, a semitone. C–D and F–G: a black key between, a tone.

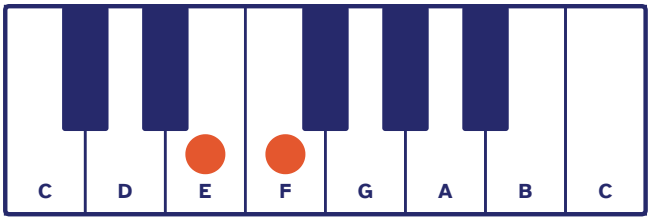
From any key to the **very next key**, black or white, is a **semitone**: the smallest step on the piano. Two semitones make a **tone**. On the white keys most steps are tones, but **E–F** and **B–C** have no black key between them: they are semitones.

THE RULE

Semitone: the very next key, with nothing between: white to white (E–F, B–C), white to black, or black to white. Tone: exactly one key between.

EXAMPLE 1

Tone or semitone?

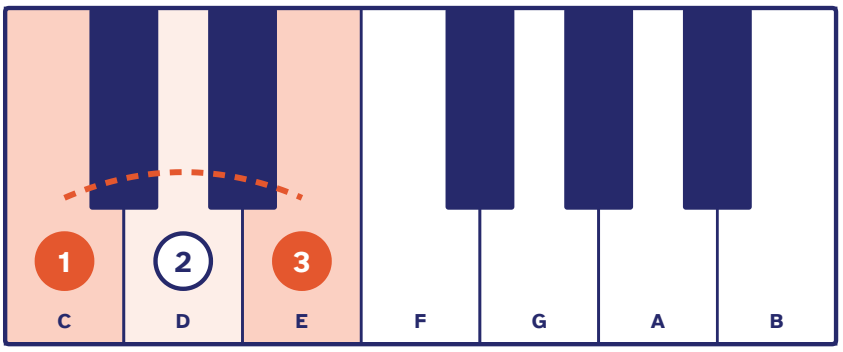


1. **Look between the red keys:** is there a black key?
2. **No black key** means they are next-door keys.
3. **Now try C and D**: is there a black key between?

E–F is a **semitone** (next door). C–D has C♯ between: a **tone**.

PART I · PITCH · 7 · CONTINUED

Intervals: count the notes



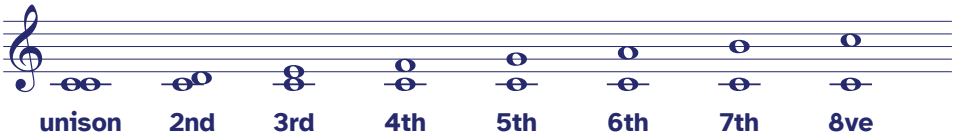
C to E: count the notes, 1 2 3: a 3rd

THE RULE

Count the letters for the number: start at 1 on the first note and count white keys, or lines and spaces, never the black keys. C to E is C D E: 3 notes, a 3rd. Sharps and flats never change the number.

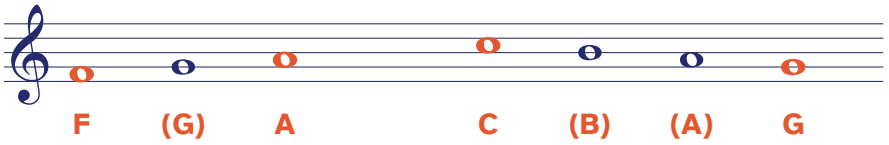
A step on the staff (the next letter) may be a tone (C–D) or a semitone (E–F); C to C♯ is a semitone with no new letter.

Interval numbers from C



EXAMPLE 2

How far apart are these notes?



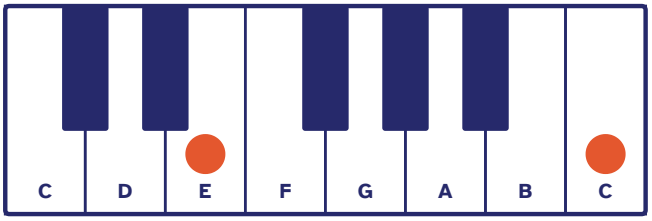
1. **Call the first note 1:** here F is 1.
2. **Count every letter** up to the second note.
3. **Going down works the same:** start at 1 on the top note and count down.

F G A is 3 notes: a **3rd**. C B A G is 4 notes: a **4th**.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 3

Name an interval on the keyboard

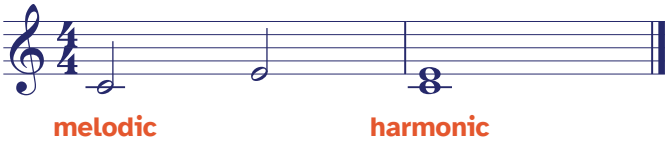


1. **Find both red keys:** E, and the C above it.
2. **Count the notes** from E: E is 1 …
3. **Stop on the C.** What number is it?

E F G A B C is 6 notes: a **6th**.

EXAMPLE 4

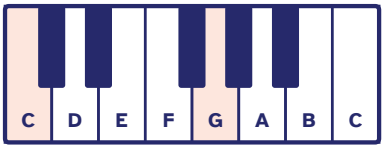
Melodic or harmonic?



1. **The first bar:** C, then E, one after the other.
2. **The second bar:** C and E at the same time.
3. **Count the notes** in each.

Both are a **3rd**: **melodic** when played one after the other, **harmonic** when played together.

TRY IT · 10 SECONDS



Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



"Twinkle, twinkle": C up to G, C D E F G, **five notes, a 5th**.

Try it. Sing the first two notes of a song and count them, both included.

TRAP

× Counting only the gaps: C to E is 3 notes, not 2 gaps. And a sharp or flat never changes the number: C to G♯ is still a 5th.

SEE ALSO
Major, minor and perfect p. 45 · Major scales p. 93

PRACTISE IT
Workbook Pitch 7 · Online: Intervals · Tones & Semitones

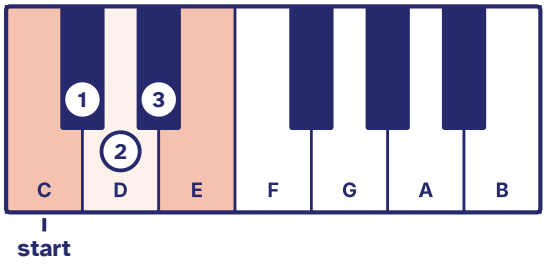
PART I · PITCH · 8

Major, minor and perfect

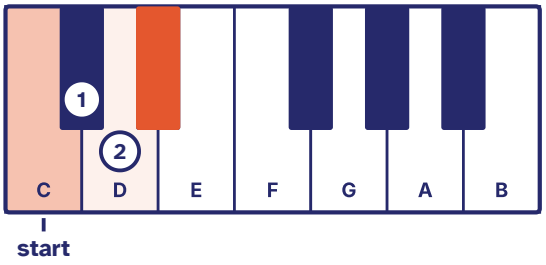
C to E and C to E♭ are both 3rds: three notes each. But they sound different, bright and dark. To tell them apart, **count the keys in between**, black and white: the keys you do not play.

THE SECRET

Count the keys between: 3 make a major 3rd, 2 a minor 3rd.



C to **E**: 3 keys between. A **major** 3rd: bright.



C to **E♭**: 2 keys between, one fewer. A **minor** 3rd: dark.

One key fewer between: a major 3rd becomes minor

THE RULE

Count the notes for the number (a 3rd). Count the keys in between for the size (major or minor): black and white, not the two you play.

Number: count the letters, both ends included. **Size**: count every key in between, black and white, leaving both ends out.

Why count the keys in between? Because the size is what you hear: one key more or less turns bright into dark, or makes a 5th sound hollow and strange.

NUMBER	KEYS BETWEEN: SMALLER	MAIN SIZE
3rd	2 (minor)	3 (major)
4th		4 (perfect)
5th		6 (perfect)
6th	7 (minor)	8 (major)
octave		11 (perfect)

3rds and 6ths come in two sizes: **major**, and **minor**, one key fewer between.

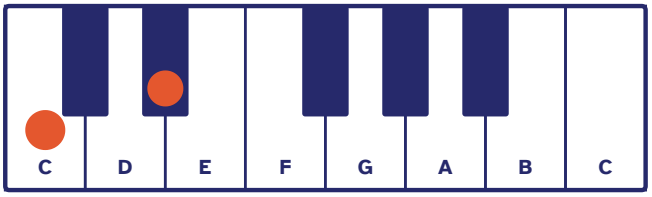
4ths, 5ths and octaves have one main size, called **perfect**. They are never major or minor.

2nds and 7ths have two sizes too (p. 93). F up to B has 5 keys between: not perfect; its name comes later. Semitones = keys between + 1: a major 3rd is 4.

PART I · PITCH · 8 · CONTINUED

EXAMPLE 1

Major or minor 3rd?

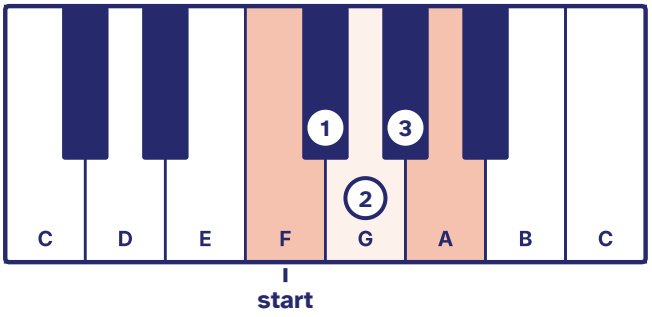


1. **Count the notes:** C D E, three letters, so it is a 3rd.
2. **Count the keys in between** C and the red black key, E♭: C♯, D …
3. **A major 3rd has 3 between.** How many here?

C to E♭ has **2 keys between**: a **minor 3rd**.

EXAMPLE 2

Name F up to A

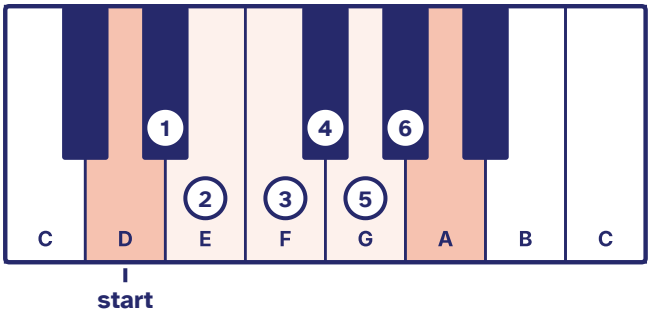


1. **Count the notes:** F G A is 3, so it is a 3rd.
2. **Count the keys in between:** they are numbered.
3. **3 between, or 2?**

F to A has 3 keys between: a **major 3rd**, the same size as C to E.

EXAMPLE 3

Name D up to A



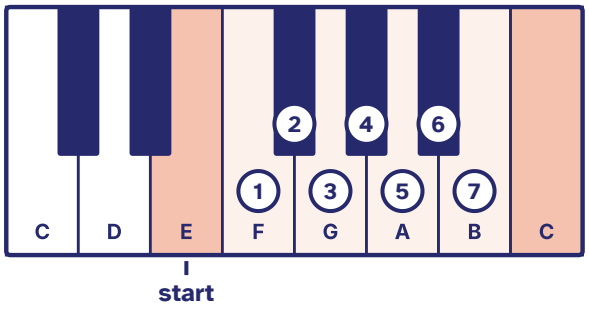
1. **The number:** D E F G A, five notes, a 5th.
2. **The size:** count the keys in between: the numbered keys.
3. **A 5th with 6 between** is its main size. What is that called?

D up to A is a **perfect 5th**: 6 keys between.

PART I · PITCH · 8 · CONTINUED

EXAMPLE 4

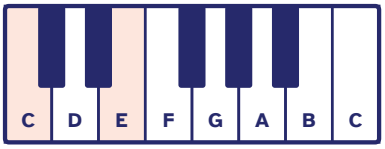
Name E up to C



1. **The number:** E F G A B C, six notes, a 6th.
2. **The size:** count the keys in between.
3. **A major 6th** has 8 between. Is this one more or one fewer?

E up to C has 7 keys between, one fewer than major: a **minor 6th**.

TRY IT · 10 SECONDS



Play C–E, then C–E♭.

Bright, then dark. One key makes the difference: the major 3rd has three keys between, the minor 3rd two.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to E♭ is three notes, a 3rd, with 3 keys between (F♯, F, E): a **major 3rd**. Counting down works the same way; bright and dark is clearest when you play the two 3rds side by side.

Try it. Count the keys between G and E♭ on the keyboard.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key between, black and white.
- × Counting the two notes you play: count only the keys in between.
- × Calling a 4th or 5th “major”: they are perfect.
- × C–D♯ is a 2nd, C–E♭ a 3rd: the same keys, but the letters decide the number.

SEE ALSO
Counting intervals p. 42 · Chords and triads p. 110

PRACTISE IT
Workbook Pitch 8 · Online: Intervals (Major or minor 3rd)

PART I · PITCH · TIPS

When a note reads wrong

Most wrong notes come from a handful of slips. Name the slip first, then fix only that note: playing the whole line again rarely finds it.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
The left hand reads like the right	**the clef**: bass lines are G B D F A, not E G B D F	p. 32
You land an octave too high or too low	**the Cs**: find Middle C first, then walk out to Bass C or Treble C	p. 23
Notes above or below the staff slow you down	**the ledger lines**: step on from the nearest staff line	p. 36
A sharp or flat vanishes, or stays too long	**the bar line**: a sign lasts only to the end of its bar	p. 40
You play the black key on the wrong side	**the direction**: sharp goes right, flat goes left	p. 39
Two notes sound too close or too far apart	**the interval**: count the notes, then the keys between	p. 45

EXAMPLE

Spot the slip



?

Sam played A. Wrong!

1. **Check the clef first:** this is the bass clef, not the treble.
2. **Use the bass sentence:** the spaces are *All Cows Eat Grass*.
3. **The red note is in space 2:** the second word.

The note is **C** (*Cows*): Bass C. Sam read the bass staff with treble names (treble space 2 is A): the most common slip of all.

HOW TO

Three checks before you read a new line

1. **The clefs:** which staff is treble, which is bass?
2. **The clef notes:** find treble G, bass F and Middle C, and read the other notes from them.
3. **The signs:** mark each sharp, flat or natural, and see where its bar ends.

THINK OF IT Like reading a map: find the big landmarks first, then the streets around them. A note far from every clef note is the one most often misread.